# Andrzej Pokraka

## Personal information

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## Academic appointments

2025–30 **Senior Postdoctoral Researcher (5 yr), Universe+ ERC Synergy Grant**
University of Amsterdam, Netherlands (2025–27)
Institute for Advanced Study, Princeton, USA (2027–30)

2022–25 **Postdoctoral Researcher**
Brown University, Providence, USA

## Education

2022 **Doctor of Philosophy in Physics**
McGill University, Montreal, Canada • Advisor: Prof. Simon Caron-Huot

2017 **Master of Science in Physics**
University of Alberta, Edmonton, Canada • Advisor: Prof. Andrzej Czarnecki

2015 **Bachelor of Science in Physics, High Honours**
University of Saskatchewan, Saskatoon, Canada • Minors in Mathematics and Studio Art

## Awards and fellowships

2019–22 Fonds de recherche Nature et technologies Québec doctoral award
CAD 80,000 • Main applicant • McGill University

2017–19 Walter Sumner Foundation Fellowship
CAD 16,000 • McGill University

2016–17 NSERC Canada Graduate Scholarship – Master's
CAD 17,500 • Main applicant • University of Alberta

2015–16 University of Alberta Graduate Recruitment Scholarship
CAD 22,500 • University of Alberta

2015–16 Walter H. Johns Graduate Fellowship
CAD 5,000 • University of Alberta

2013, 2014, 2015 NSERC Undergraduate Student Research Award
3 × CAD 4,500 • Co-applicant • University of Saskatchewan

## Research

I am a mathematical physicist trained in flat-space perturbative quantum field theory. In my doctoral work with Simon Caron-Huot, I built relative twisted (co)homology for Feynman integrals. As a postdoc at Brown, I extended it to the higher-genus geometries of collider and gravitational-wave physics and to the double copy of string theory. Since 2023, I have carried the same tools into cosmology, where they explained structures that had been found empirically: the physical basis of correlators from cuts, kinematic flow from the flow of cuts, and a coaction for FRW correlators.

## Publications

As is customary in my field, authors are listed alphabetically. Full list on INSPIRE.

[1] C. Larkin, A. J. McLeod, A. Pokraka and L. Ren, *Direct-to-Symbol Integration from Landau Analysis*, 2609.15952.

[2] A. J. McLeod, A. Pokraka and L. Ren, *A Graphical Coaction for FRW Integrals from Partial/Relative Twisted (Co)homology*, 2606.13627.

[3] G. Crisanti, H. Frellesvig, A. Pokraka and S. Smith, *Magic Relations and Critical Varieties of Feynman Integrals*, 2605.29789.

[4] A. McLeod, A. Pokraka and L. Ren, *A Graphical Coaction for FRW Wavefunction Coefficients*, 2603.25703.

[5] J. Gong, A. Pokraka, K. Yan and X. Zhang, *Toward the analytic bootstrap of energy correlators*, *JHEP* **06** (2026) 140, [2509.22782].

[6] A. Pokraka, L. Ren and C. Rodriguez, *A double copy from twisted (co)homology at genus g*, 2509.01598.

[7] R. Glew and A. Pokraka, *Kinematic flow from the flow of cuts*, *JHEP* **06** (2026) 158, [2508.11568].

[8] A. Pokraka, M. Spradlin, A. Volovich and H.-C. Weng, *Symbol alphabets in QCD and flag cluster algebras*, *JHEP* **03** (2026) 120, [2506.11895].

[9] S. De, S. Paranjape, A. Pokraka, M. Spradlin and A. Volovich, *Hidden zeros of the cosmological wavefunction*, *JHEP* **07** (2025) 174, [2503.23579].

[10] S. De and A. Pokraka, *A physical basis for cosmological correlators from cuts*, *JHEP* **03** (2025) 040, [2411.09695].

[11] S. De, A. Pokraka, M. Skowronek, M. Spradlin and A. Volovich, *Surfaceology for colored Yukawa theory*, *JHEP* **09** (2024) 160, [2406.04411].

[12] A. Pokraka, S. Rajan, L. Ren, A. Volovich and W. W. Zhao, *Five-dimensional spinor helicity for all masses and spins*, *JHEP* **07** (2025) 056, [2405.09533].

[13] M. Giroux, A. Pokraka, F. Porkert and Y. Sohnle, *The soaring kite: a tale of two punctured tori*, *JHEP* **05** (2024) 239, [2401.14307].

[14] R. Bhardwaj, A. Pokraka, L. Ren and C. Rodriguez, *A double copy from twisted (co)homology at genus one*, *JHEP* **07** (2024) 040, [2312.02148].

[15] S. Caron-Huot, A. Pokraka and Z. Zahraee, *Two-point sum-rules in three-dimensional Yang-Mills theory*, *JHEP* **01** (2024) 195, [2309.04472].

[16] M. Giroux, A. Pokraka, F. Porkert and Y. Sohnle, *Looping the loops: a tale of elliptic dual Feynman integrals*, *PoS* **RADCOR2023** (2024) 033, [2309.04592].

[17] S. De and A. Pokraka, *Cosmology meets cohomology*, *JHEP* **03** (2024) 156, [2308.03753].

[18] M. Giroux and A. Pokraka, *Loop-by-loop differential equations for dual (elliptic) Feynman integrals*, *JHEP* **03** (2023) 155, [2210.09898].

[19] S. Caron-Huot and A. Pokraka, *Duals of Feynman Integrals. Part II. Generalized unitarity*, *JHEP* **04** (2022) 078, [2112.00055].

[20] S. Caron-Huot and A. Pokraka, *Duals of Feynman integrals. Part I. Differential equations*, *JHEP* **12** (2021) 045, [2104.06898].

[21] R. Gonzo and A. Pokraka, *Light-ray operators, detectors and gravitational event shapes*, *JHEP* **05** (2021) 015, [2012.01406].

[22] S. Mizera and A. Pokraka, *From Infinity to Four Dimensions: Higher Residue Pairings and Feynman Integrals*, *JHEP* **02** (2020) 159, [1910.11852].

[23] A. H. Fariborz, J. Ho, A. Pokraka and T. G. Steele, *The Bridge Between Chiral Lagrangians and QCD Sum-Rules*, *Nucl. Part. Phys. Proc.* **309-311** (2020) 119–123, [1909.07243].

[24] A. Pokraka and A. Czarnecki, *Parapositronium can decay into three photons*, *Phys. Rev. D* **96** (2017) 093002, [1707.09466].

[25] A. Pokraka and A. Czarnecki, *Positronium Decay into a Photon and Neutrinos*, *Phys. Rev. D* **94** (2016) 113012, [1611.07645].

[26] A. Pokraka and R. Dick, *Dimensional Effects on the Density of States in Systems with Quasi-Relativistic Dispersion Relations and Potential Wells*, 1511.09421.

[27] A. H. Fariborz, A. Pokraka and T. G. Steele, *Connections between chiral Lagrangians and QCD sum-rules*, *Mod. Phys. Lett. A* **31** (2016) 1650023, [1505.05553].

## Conference presentations

Sep 2026 *A coaction for FRW correlators from relative/partial twisted (co)homology*
Scattering Amplitudes: from Positive Geometries to Analytic Structures • Pollica Physics Center, Pollica, Italy

Sep 2026 *A double copy at higher genus from twisted (co)homology*
Elliptics and Beyond '26 • SwissMAP Research Station, Les Diablerets, Switzerland

Jun 2026 *Kinematic flow and new languages for correlators*
Expanding Thoughts on de Sitter • La Ricotta, As Barreiras, Spain

Mar 2026 *A diagrammatic coaction for FRW power-law cosmological correlators* • recorded
Amplitudes and Algebraic Geometry • Erwin Schrödinger Institute, Vienna, Austria

Jan 2026 *Twisted (co)homology and single-valued integration*
YOUNGST@RS: Single-Valued Periods in Scattering Amplitudes (virtual) • Mainz Institute for Theoretical Physics, Germany

Nov 2025 *Relative twisted (co)homology for amplitudes and cosmology*
YOUNGST@RS: Loop-the-Loop (virtual) • Mainz Institute for Theoretical Physics, Germany

Nov 2025 *Zonotopes in cosmology: where and how to find them*
Cosmology Meets Non-Linear Algebra • University of Amsterdam, Netherlands

Oct 2025 *Relative twisted (co)homology in QFT*
Michigan Math-Physics Amplitudes Workshop • University of Michigan, Ann Arbor, USA

Sep 2025 *Relative twisted (co)homology in QFT*
New Ways To Higher Points • Humboldt University of Berlin, Germany

Sep 2024 *A double copy from twisted (co)homology at genus-one*
Eurostrings 2024 • University of Southampton, UK

Aug 2024 *Surfaceology for colored Yukawa theory* • recorded
The Mathematics behind Scattering Amplitudes • Galileo Galilei Institute, Florence, Italy

Aug 2024 *A physical basis for cosmological correlators from cuts*
Special Functions: from Geometry to Fundamental Interactions • TU Munich, Germany

Jul 2024 *Kinematic flow from the flow of cuts*
Amplitudes Afterparty • National Taiwan University, Taipei, Taiwan

Jun 2024 *Two-point sum-rules in three-dimensional Yang-Mills theory*
S-Matrix Bootstrap • Reykjavík University, Iceland

Jun 2024 *An epsilon-form differential equation for the 5-mass kite: a tale of two elliptic curves*
Beyond Elliptic Polylogarithms • Casa Matemática Oaxaca, Mexico

Mar 2024 *A double copy from twisted (co)homology at genus-one*
*S*-matrix Marathon • Institute for Advanced Study, Princeton, USA

Feb 2024 *Cosmology meets cohomology*
Universe+ Kickoff Conference • MPI for Mathematics in the Sciences, Leipzig, Germany

Sep 2023 *Cosmology meets cohomology*
Mathemamplitudes 2023 • University of Padova, Italy

Sep 2023 *A double copy from twisted (co)homology at genus-one*
Elliptics and Beyond • ETH Zürich, Switzerland

Dec 2022 *Loop-by-loop differential equations for dual (elliptic) Feynman integrals*
QCD Meets Gravity • ETH Zürich, Switzerland

Aug 2022 *The Duals of Feynman Integrals*
Amplitudes 2022 • Charles University, Prague, Czech Republic

May 2022 *The Duals of Feynman Integrals*
Loop Fest XX • University of Pittsburgh, USA

Dec 2021 *The Duals of Feynman Integrals*
QCD Meets Gravity (virtual) • University of California, Los Angeles, USA

Oct 2021 *Duals of Feynman Integrals, II: Generalized Unitarity*
Elliptics 2021 (virtual) • Uppsala University, Sweden

Jun 2021 *Duals of Feynman Integrals, I: Differential Equations* • recorded
CAP Virtual Congress • Canadian Association of Physicists

Jul 2019 *On the Poincaré dual of Feynman integrals*
Amplitudes 2019 (poster) • Trinity College Dublin, Ireland

Jul 2016 *Positronium decay into a photon and neutrinos*
Women in Physics Canada Conference • University of Saskatchewan, Saskatoon, Canada

Feb 2016 *New positronium decay channel*
Winter Nuclear and Particle Physics Conference • TRIUMF, Banff, Canada

## Invited seminars

Dublin Area Theoretical Physics Colloquium, Trinity College Dublin (2026) • Durham (2026) • Joint Belgium High-Energy Seminars, Ghent (2025) • Universe+ Synergy Grant series: IAS Princeton, Amsterdam, MPP Munich, MPI MiS Leipzig (2025) • Perimeter Institute (2025) • MaScAmp Synergy Grant series: MPI for Gravitational Physics, Trinity College Dublin, Uppsala and Oxford (2025) • Uppsala (2025, 2023) • McGill (2024) • Institute for Advanced Study (2024, 2022, 2021) • Bonn (2023) • Centre de Recherches Mathématiques, Montreal (2022) • Hertfordshire (2022, 2021) • Freiburg (2021) • Turin (2021) • CERN (2021) • Michigan State (2019).

### Maintained software

2026 `frwcoaction.ca` • Web app for the graphical coaction of FRW correlators (source)

2026 `FRWcoaction_Mathematica` • Mathematica notebooks for 2603.25703 and 2606.13627

2025 `KinematicFlowFromCuts` • Mathematica code for 2508.11568 (with R. Glew)

2024 `PhysicalBasisFromCuts` • Mathematica notebooks for 2411.09695 (with S. De)

## Service, mentoring and teaching

### Service and organizing

2026– Organizer • Universe+ Online Seminar Series • University of Amsterdam

Nov 2025 Organizer (3 days, ∼30 participants) • Cosmology Meets Non-Linear Algebra • University of Amsterdam

Apr 2023 Organizer (1 day, ∼80 participants) • New England Strings 2023 • Brown University

Sep 2016 Organizer • Graduate Student Physics Research Symposium • University of Alberta

ongoing Journal referee: JHEP (since 2019) • Phys. Rev. D (since 2023)

### Mentoring

I have mentored 10 junior researchers (9 PhD students and one undergraduate) during my four years as a postdoc. Each project with these students has resulted in a paper. In 5 of these projects, I was the senior researcher responsible for the initial ideas and for guiding the students through their computational and conceptual roadblocks.

PhD students mentored:

- M. Giroux (McGill University, now a postdoc at Columbia University and the Institute for Advanced Study). Differential equations of elliptic Feynman integrals.
- L. Ren (Brown University, now a postdoc at Queen Mary University of London). Higher-dimensional spinor helicity and the higher-genus double copy.
- S. De (Brown University, now a postdoc at the University of Pennsylvania and the Institute for Advanced Study). Differential equations for FRW cosmological correlators.
- R. Bhardwaj (Brown University). Higher-genus double copy via twisted (co)homology.
- M. Skowronek (Brown University). Extending the surfaceology formalism to fermionic amplitudes.
- F. Porkert (Bonn University). Differential equations of elliptic Feynman integrals.
- C. Rodriguez (Uppsala University, now a postdoc at the Max Planck Institute for Mathematics in the Sciences, Leipzig, moving to Okinawa). Higher-genus double copy via twisted (co)homology.
- Y. Sohnle (Uppsala University, now a joint postdoc at SLAC and DESY). Differential equations of elliptic Feynman integrals.
- C. Larkin (University of Edinburgh). Direct-to-symbol integration from Landau analysis.

Undergraduate students mentored:

- S. Rajan (Brown University, now a PhD student at Berkeley). Higher-dimensional spinor helicity formalism.

### Teaching

Guest lectures during my PhD and postdoctoral appointments: in graduate electromagnetism (Jackson), graduate quantum field theory (Srednicki) and fourth-year quantum mechanics (Griffiths)

2017–19 Teaching assistant • McGill University • Electrodynamics I and II (Griffiths); advanced quantum mechanics

2015–18 Organized preparation sessions for the CAP University Prize Exam • Alberta (2015, 2016), McGill (2017, 2018) • my students placed 5th (2016) and 9th (2017) in Canada

2015–17 Teaching assistant • University of Alberta • First- and second-year physics help desk